

GUIDE TO UNDERSTANDING TRICORDER/OS

M-TOR — Multi-Tronic Operating Realm for Android

A theory-of-operations paper for developers and scientists

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This document describes software that runs on an ordinary phone. It is not an operating system in the Linux or Android-kernel sense, not DO-178C airborne software, and not affiliated with Star Trek rights holders. “Realm” here means a coordination layer that sits on Android and treats models, sensors, and radios as swappable organs.

1. What M-TOR is

Tricorder/OS is not a third Android. The phone still boots Samsung’s kernel and ART. M-TOR is the multi-tronic operating realm: a thin, event-shaped control plane that makes several heterogeneous computers on one SoC behave like one instrument.

Those computers are real and separate. Gemma and SmolVLM live in Termux llama-server processes on loopback. Stable Diffusion is another process. Kokoro is another. YAMNet, BirdNET, and flora/fauna are in-process TFLite inside the APK. Speech-in is Android’s SpeechRecognizer. Comlink is Google Nearby. The face is a WebView. Kotlin is the janitor that owns hardware and permissions.

An app would pick one of those and wrap it. An integration treats them as organs. M-TOR is the circulatory system.

The name is deliberate. Multi-tronic: more than one inference engine, more than one sensor class, more than one transport, none allowed to become the system. Operating realm: a jurisdiction with laws (tuples, event names, JSON row shape), not a new kernel.

2. Design laws

Five laws were written before the first APK, in the v0.1 design spec, and they still bind the shipped 0.6.x face.

Law 1. Nothing important is a special case in application code. Modes, keyword hints, ports, and providers are data. Adding flora did not require a new Android Activity. Adding Comlink did not require a new LLM.

Law 2. API-first. The face never owns a model. It posts JSON to 127.0.0.1 or it calls a JavascriptInterface that already looks like a tool. Swap Gemma for another GGUF without touching the CRT.

Law 3. Event-driven. Nobody important calls anybody important directly. A button, a frame, a bird label, a PTT packet, and a NOAA row all become events or event-shaped JSON. Subscribers decide whether they care.

Law 4. Algebraic intent. User meaning is not inferred by an LLM at the door. It is projected onto a tuple $I = (\text{Mode}, \text{InputChannel}, \text{TargetModality}, \text{Action})$ and matched against ordered rules. The model is a worker hired after the route exists.

Law 5. Extensible by construction. A new sense is a provider with the same emit shape as NOAA already had. If it cannot be expressed as a row {kind, text, from, guid, hmmm, extra}, it does not belong on the bus yet.

3. The two spines that became one realm

3.1 tricorder-core (the algebraic spine)

The Node package tricorder-core is the written constitution. EventBus: the only object UI, providers, and engines talk to. TupleEngine: deterministic router, zero domain literals. payloadBuilder and mcpClient: turn a resolved route into a backend call. sensorLoopDriver and contextProviderPoller: time, not UI, owns Sensor and Sentry and weather. Tests run with no network.

That spine still matters even though the Fold 3 face grew a second implementation. The laws did not change. The Android build re-hosted them where latency and permissions live.

3.2 Android hosting (the visceral spine)

WebView renders phosphor. voice-menu.js and submitInquiry() are the tuple door: normalize STT, intercept COMPUTER / CLOCK / COMLINK / NAVIGATE TO before Gemma. Kotlin owns camera, mic, GNSS, baro, Nearby, TFLite, MediaPlayer. Termux keeps four warm servers. FieldLog and AudioLog are the durable riverbed.

This split is why new capabilities do not feel like “another app inside the app.” A scientist adding a classifier does not open a new Activity stack. They emit the same row the CRT already knows how to paint and the mesh already knows how to forward.

4. Intent as water

Intent is not a prompt. It is a drop that must fall downhill through weirs until it reaches a worker or a drain. Gravity is the tuple. Channels are the pipes. Reservoirs are logs. Weirs are Hold, MUTE, IPL, and peer mute.

```
utterance / button / frame / PCM / ADS-B
  | normalize (STT aliases, punctuation strip)
  v
tuple door  I = (mode, channel, modality, action)
  | first matching rule wins – no LLM
  +--> local organ      (clock, comlink, dictation, NAV)
  +--> in-process net   (YAMNet, flora, BirdNET)
  +--> loopback organ   (Gemma, SmolVLM, SD, Kokoro)
  +--> radio organ      (Nearby JSON)
  v
event-shaped row
  | CRT subscriber, LED subscriber, log subscriber,
  | TTS subscriber, mesh subscriber
  v
field log + optional spoken line + optional remote copy
```

Water does not ask the ocean what a creek is. A spoken “tricolor commlink” is aliased to the comlink action before any model token is spent. That is the difference between an orchestrator and a chatbot with plugins.

When the drop does reach Gemma, the expensive work is isolated on :8080. The face stays interactive. IPL / ALL is a weir that dumps the cistern: in-flight HTTP and TTS die without tearing down the realm.

5. Why stacking does not stall the realm

“No system lag” is not a claim that Stable Diffusion is instant. It is a claim that a new organ does not add a synchronous frame to the UI thread or a new token to the door.

5.1 Warm processes, cold features

The four Termux servers stay up. A feature flag or a button does not fork a 4B model. Flora fetch writes a file; the next Sensor frame already knows the extra classifier. Comlink starts advertising on page-finished; PTT does not spin up a SIP stack.

5.2 Cheap door, expensive room

Tuple matching is string work. YAMNet is milliseconds on the 888. Gemma is seconds. Kokoro is seconds. SD is tens of seconds. The realm stays snappy because the door never waits in the expensive room. Hold on Sensor/Sentry is a weir: one frame, one describe, stop the flood. Continuous machine vision writes the CRT and skips Kokoro. MUTE kills every speaker and leaves the river of rows intact.

5.3 Same row, new subscriber

Gunfire on unit Spock is not a new protocol. It is `emitContextAlert(local=true)` which already logs, paints, speaks, and, if Comlink is live, sends `{kind:context, from, guid, source, text, hmmm}`. Kirk's `handleComlink` is a subscriber. Adding “prepend the callsign” was a field on the row, not a new bus.

5.4 Abort is a first-class organ

IPL exists because stacking fails if work cannot be cancelled. A scientist who adds a 30-second image job without an abort has coupled the face to that job. M-TOR refuses that coupling.

5.5 What we do not do

We do not run a second Android Activity per sense. We do not ask Gemma “the user said something, what now?” We do not put Dendrite in Termux so two Folds can miss each other in a valley and pretend that is a mesh. We do not block `onCreate` on model load.

6. Data as water

A field-log row is the molecule. Everything that matters can be poured into one: kind, text, timestamp, optional photo path, optional extra object (tail, plus code, from, guid). AudioLog is the same idea for YAMNet dwells. Dictation notes can be poured into the field log with LOG nnnn.

Readings charts are not a second database. They are a filter over that river: by kind, by day/week/month, by tail, by airframe. Touching bar A is a downstream sip, not a join across apps.

Comlink is the same molecule on a radio. `Payload.fromBytes(row.toString())`. There is no separate “alert API” and “chat API.” PTT is `kind=ptt`. A watch is `kind=context`. Hello is `kind=hello`. If a future Earth-link store-and-forward appears, it should ingest that molecule, not invent a homeserver schema.

Place names flow the same way. A fix becomes lat/lon plus an offline plus code plus optional Geocoder city. NAV, horizon, and log all drink from one cache. That is why “Newburgh” appears on the horizon without a navigation SDK.

7. Theory of operation by organ

7.1 Inquiry

STT or typed line hits `submitInquiry`. Wakes (dictation, diagnostics, `commlink`, clock, TRICORDER ...) leave. Else POST :8080. CRT streams tokens. Kokoro speaks in ≤80-character chunks unless the length weir is off. COMPUTER is a scripted easter-egg, not a model.

7.2 Sensor / Sentry

A frame is a payload, not a conversation. SmolVLM describes. Flora TFLite may overlay a species bar. Sentry does not wait for motion to caption; motion is one more label. Keyword watches fire robot voice and a log row. Hold is the weir against 3-second redescribe.

7.3 Audio

Duty cycle (dwell/interval sliders) is the only clock that organ needs. PCM → YAMNet → optional BirdNET → bucket fusion → TOP 3 bars → watch rules. The unit's own TTS is not classified as HUMANS DETECTED. Context providers (NOAA, OpenSky) emit on the same alert pipe.

7.4 Imagine

A prompt is a job on :8188. Steps and size are operator weirs on a turbo checkpoint, not hidden magic. The face does not freeze; IMAGE lamp goes amber. Result is a bitmap plus a spoken one-liner.

7.5 Translate / knowledge / clock / NAV

Each is a local organ behind a tuple. Translator downloads packs once. MEMORY ALPHA and MEDICAL are search organs with TERMINATE as the drain. CLOCK drinks GnsClock when it can. NAV drinks the same fix cache and

writes CAMP/DEST as ordinary points, not a routing graph.

7.6 Comlink

Advertising name CALLSIGN|GUID. P2P_CLUSTER. Smaller GUID requests. Auto-accept. Roster is state on peers, not a chat server. Mute is a receive-side weir. Robot voice only, so a gunfire line from the tent does not wait on Kokoro.

8. How a scientist adds an organ

Write a producer that can emit the molecule. If it is PCM-in, classify in-process and append AudioLog / FieldLog. If it is a cloud or radio feed, look like contextProviderPoller: interval, normalize, emit. If it is an operator verb, add a tuple alias at the door (and the STT misspellings). If other units should hear it, put from+guid on the row and send() when Comlink.running. Paint with an existing CRT subscriber or a ten-line sheet. Do not add a mode that only an LLM can enter.

The test of belonging: can a deaf, muted, offline unit still write the row? If yes, the organ is in the realm. If the feature dies when Gemma is down, it was an app plugin, not an organ.

9. Why this is unusual

Phone “AI” products in 2026 are almost all one model plus a skin, or a cloud round-trip with an on-device teaser. Pocket-class apps host a chat GGUF. Camera apps host a vision GGUF. None of them treat watch, flora, ADS-B, dictation, walking NAV, and a mesh of callsigns as the same river.

Cloud vendors could ship this appliance. They optimize for tokens billed and models upgraded. An offline orchestrator with four local servers and a tuple door does not sell seats. That is an economic fact, not a technical mystery. The Fold 3 already had the SoC. M-TOR is permission to use all of it at once without asking a frontier model what a button means.

10. Limits (read these)

The WebView face is one JavaScript heap. A syntax error in paintNav takes the whole CRT down; 0.6.42 did that. Nearby is Play-services radio range, not a valley-proof log sync. TFLite packs can fail closed. Termux processes die when the phone sleeps hard; ~/bin/tricorder is the babysitter, not the kernel. “No lag” does not apply to SD or first-token Gemma. Do not fly with the horizon.

11. Short glossary

Tuple — I = (Mode, InputChannel, TargetModality, Action). Door, not model.

Organ — a process or in-process classifier with one job and a row-shaped output.

Molecule — JSON row the log, CRT, TTS, and mesh can all drink.

Weir — Hold, MUTE, IPL, peer mute, TTS length. Flow control without shutting the river.

Realm — the jurisdiction those objects share on top of Android.

12. Document control

Companion to Field Manual Rev 0.6.47 and design spec v0.1. Software described: Project Tricorder Mk I 0.6.47. Source: tricorder-core (EventBus, TupleEngine) and android-app (viewport + organs).